

UPSC PRELIMS 2026: ESSENTIAL TECH TERMINOLOGY

SECTION (1): MEMFLATION (Memory Inflation)

Term coined by Gartner: Rapid cost & demand inflation for memory chips (e.g., HBM) driven by AI.



DRIVERS: GenAI Boom, Large Language Models (LLMs).

CONSEQUENCE: Global shortage of High Bandwidth Memory.



IMPACT:

AI Infrastructure Bottleneck, Data Center Cost surge.

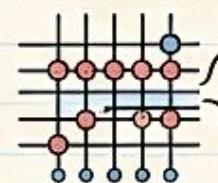


INDIA'S SEMICONDUCTOR MISSION

RAHUL BHARDWAJ
DNA SESSION: 16-04-2026

SECTION (2): MEMRISTOR & NEUROMORPHIC COMPUTING

► The 4th Fundamental Circuit Element (Resistor, Capacitor, Inductor... **MEMRISTOR**).



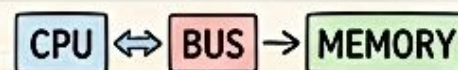
Crossbar Array: Hafnium Oxide (HfO_2) layer switching resistance.

Remembers state even with power **OFF**.

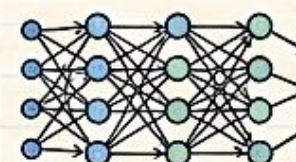
VON NEUMANN

vs.

NEUROMORPHIC



Energy & Time Waste



PROCESS & STORE IN-PLACE
SYNAPTIC EMULATION

► **BRAIN-INSPIRED SOLUTION:**
Mimics human synapses, Massive Parallel Processing, Ultra-low Power.

► **HISTORY:** Leon Chua (Hypothesized 1971)
→ HP Labs (Realized 2008).

KEY ADVANTAGES:
High Density,
Scalability,
Faster Processing.

ABHISHA AI
BY ABHISHEK PATHAK

(1) CARBON CAPTURE AND STORAGE (CCS): MINERAL CARBONATION

PERMANENT SEQUESTRATION via SOLID MINERALS

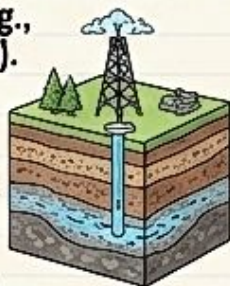
- The Concept: Reacting CO_2 with metal oxides (Ca, Mg) to form stable carbonate minerals like Calcite (CaCO_3) or Magnesite (MgCO_3).



- Comparison: **IN-SITU vs EX-SITU CARBONATION**

IN-SITU

- Location: Underground injection.
- Process: CO_2 reacts with natural reactive rocks (e.g., Basalt, Peridotite).
- Advantages: Large scale potential, direct storage.
- Example: CarbFix (Iceland).



EX-SITU

- Location: Above ground processing.
- Process: Reacting CO_2 with mined minerals or industrial wastes.
- Feedstock: Serpentine, Asbestos tailings, Steel slag.
- Advantages: Controlled environment, utilizes waste.
- Challenges: Higher cost, materials handling.



- SOLID MATERIALS for Carbonation:**



Natural Minerals
(Olivine, Serpentine)



Industrial By-products
(Fly Ash, Steel Slag)



Key Advantages of Mineral Carbonation:

- Permanent (Zero Leakage Risk)
- Environmentally Stable
- Circular Economy (Wastes to Resources)

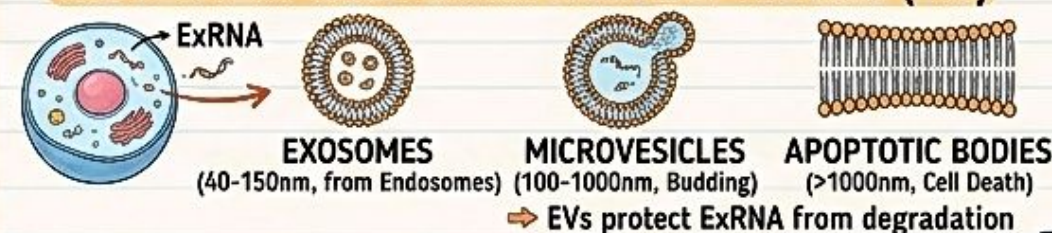
UPSC PRELIMS AND USES OF INDG RATH

(2) ExRNA ON A MISSION: EXTRA CELLULAR VESICLES



CELL-TO-CELL COMMUNICATION & APPLICATIONS

- What is ExRNA?
Extracellular RNA: RNAs found outside the cell, stable in body fluids.
- THE DELIVERY VANS: EXTRA CELLULAR VESICLES (EVs)

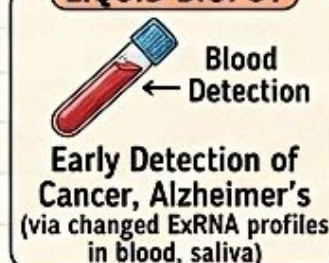


- TISSUE REPAIR AND USES OF ExRNA

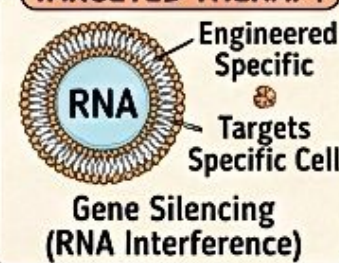


- Key Clinical Applications:

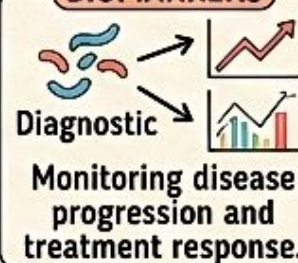
LIQUID BIOPSY



TARGETED THERAPY



BIOMARKERS



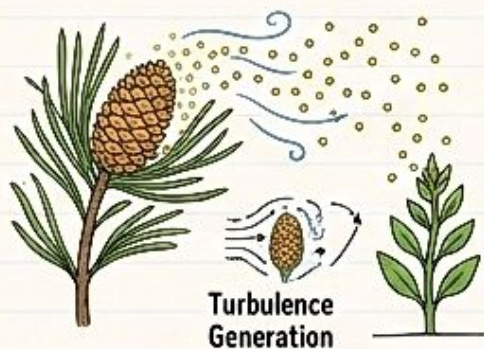
UPSC PRELIMS IMPOTANT POINTS:

- ExRNA Communication Consortium (ERCC)
- Significance in Personalized Medicine

UPSC PRELIMS 2026: KEY TOPICS - REVISION NOTES

Rahul bhardwaj DNA session 16-04-2026

1. DF-PIBM (Dynamic Floral-Pollen Interaction Biomechanical Model): WIND-INDUCED POLLEN

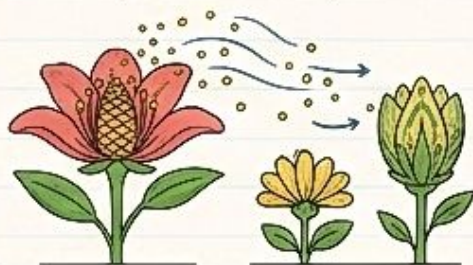


FINDINGS

- Aerodynamic Filtering:** Floral architectures (e.g., shape of pine cones, catkins) create turbulence to optimize capture.
- Vibration Resonance:** Wind causes male organs to vibrate at specific frequencies for timed pollen release.
- Settling Velocity:** Tuned grain size & density for local fluid dynamics.

FINDINGS

- Aerodynamic Filtering:** Floral architectures (e.g., shape of pine cones, catkins) create turbulence to optimize capture.
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SIGNIFICANCE

- Agriculture:** Impacts yield prediction of wind-pollinated crops (e.g., maize, wheat) with climate change.
- Biodiversity:** Understands pollen rain in fragmented forests.
- Public Health:** Aids in predicting peak 'pollen bursts' for allergy management.



2. IMD MONSOON PREDICTION (Below Normal & Drivers)



BELOW NORMAL CLASSIFICATION

When seasonal rainfall is 90% to 94% of the LONG PERIOD AVERAGE (LPA).

LONG PERIOD AVERAGE CRITERIA TABLE

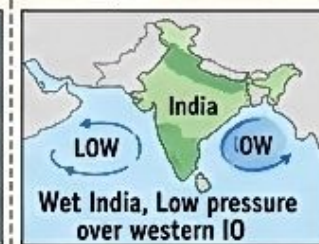
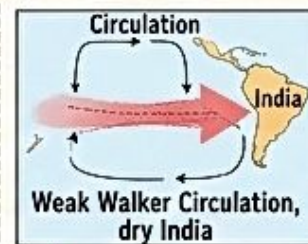
Category	Percentage of LPA
DEFICIENT	< 90%
BELOW NORMAL	90% - 94%
NORMAL	96% - 104%
ABOVE NORMAL	104% - 110%
EXCESS	> 110%

Current LPA (1971-2020 data) = 868.6 mm

EL NIÑO & IOD (The 'Tug of War')

EL NIÑO EFFECT: Abnormal warming of eastern Pacific, suppresses Indian Monsoon.

POSITIVE INDIAN OCEAN DIPOLE (IOD): Warmer western Indian Ocean, boosts Indian Monsoon, can counter El Niño.



Negative IOD exacerbates El Niño drought.

3. HUNGARY'S ELECTION & KLEPTOCRACY



KLEPTOCRACY DEFINED

Literally 'rule by thieves.' A government whose corrupt leaders use political power to appropriate nation's wealth, typically for embezzlement or misdirecting state funds.

KEY ASPECTS

- CRONY CAPITALISM:** Strategic sectors (media, construction) handed to loyalists/oligarchs.
- EROSION OF CHECKS & BALANCES:** Capturing judiciary & anti-corruption bodies to ensure impunity.
- WEAPONIZATION OF LAW:** Using legal frameworks to harass opposition & media ('Lawfare').
- TRANSNATIONAL MONEY LAUNDERING:** Use of offshore accounts, shell companies for illicit gains.
- POPULIST RHETORIC:** Masking rent-seeking behind strong nationalist/sovereignist discourse.

